# Property P for two-dimensional polyhedral Banach spaces

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## Source Question

Mandal–Manna–Paul–Sain define Property P for a Banach space $X$: there should exist $\varepsilon \in (0, 1)$ such that every bounded linear operator $T \in \mathcal{L}(X)$ preserving $\varepsilon$-approximate Birkhoff–James orthogonality at every point is a scalar multiple of an isometry [1]. Their source page contains the question and definition:

In view of the Blanco-Koldobsky-Turnšek characterization of isometries, it is natural to consider the following more general query:

*Given a Banach space $\mathbb{X}$, does there exist an $\epsilon \in (0,1)$, such that any $T \in \mathbb{L}(\mathbb{X})$ preserving $\epsilon$-orthogonality is an isometry multiplied by a constant?*

It is clear that an affirmative answer to the above query will directly lead to a proper refinement of the Blanco-Koldobsky-Turnšek theorem, which is the main motivation behind this study. In order to address the above query, it is convenient to introduce the following definition:

**Definition 1.2.** Let $\mathbb{X}$ be a Banach space. We say that $\mathbb{X}$ has the *Property P* if there exists an $\epsilon \in (0,1)$ such that the following two conditions are equivalent:
(i) $T \in \mathbb{L}(\mathbb{X})$ preserves $\epsilon$-orthogonality at each $x \in \mathbb{X}$.
(ii) $T \in \mathbb{L}(\mathbb{X})$ is a scalar multiple of an isometry.

Clearly, whenever a Banach space $\mathbb{X}$ possesses the *Property P*, a refinement of the Blanco-Koldobsky-Turnšek theorem is possible for $\mathbb{X}$. On the other hand, if $\mathbb{X}$ does not possess the *Property P*, it follows that the Blanco-Koldobsky-Turnšek theorem can not be improved by considering approximate preservation of orthogonality. Our main aim is to produce concrete examples of both types of Banach spaces by considering certain geometric properties of the concerned space. We first investigate some geometric and analytic properties of finite-dimensional polyhedral Banach spaces, including the smoothness and the number of extreme supporting functionals of each points, related to such preservation. Additionally, we analyze the linear operators satisfying such preservation of orthogonality, in connection with the geometry of the concerned spaces. As an application of this study, we characterize the isometries on some Banach spaces, thus obtaining refinements of the Blanco-Koldobsky-Turnšek theorem on these spaces. Finally, we establish that a two-dimensional Banach space, whose unit sphere is given by a regular $2n$-gon $(n > 2)$, and the sequence space $\ell_\infty^n$ $(n > 2)$ do satisfy the *Property P*. On the other hand, we prove that the sequence spaces $\ell_p$ $(1 \leq p < \infty)$ do not possess this property.

We end this section with the following characterization of the approximate Birkhoff-James orthogonality, which will be used repeatedly.

**Theorem 1.3.** [4, Th. 2.4] *Let $\mathbb{X}$ be a Banach space and let $x, y \in \mathbb{X}$. Let $\epsilon \in [0,1)$. Then $x$ is $\epsilon$-approximate Birkhoff-James orthogonal to $y$ if and only if there exists $f \in J(x)$ such that $|f(y)| \leq \epsilon \|y\|$.*

## 2. Main Results

We begin this section with the following proposition.

**Proposition 2.1.** *Let $\mathbb{X}$ be a finite-dimensional polyhedral Banach space. For each $f \in Ext\ B_{\mathbb{X}^*}$, consider the set $\mathcal{A}_f = \{x \in \mathbb{X} : J(x) = \{f\}\}$. Then the following results hold true:*

The source proves that regular $2n$-gon planes have Property P for $n > 2$, and observes that the regular 4-gon case fails because it is isometric to $\ell_1^2$. This packet removes the regularity assumption and gives the full two-dimensional polyhedral classification.

## Source Tools Used

We use three facts from [1]. First, if a nonzero operator preserves $\varepsilon$-orthogonality for some $0 \le \varepsilon < 1$, then it is injective [1, Lemma 2.7]. Second, for each finite-dimensional polyhedral space $X$ the source defines a positive constant $\varepsilon_X$ such that, for $0 \le \varepsilon < \varepsilon_X$, an $\varepsilon$-orthogonality preserver $T \in \mathcal{L}(X)$ maps nonsmooth points of $X$ to nonsmooth points of $X$ [1, Corollary 2.16]. In a polyhedral plane, the nonsmooth boundary points are exactly the vertex rays. Third, the exact Blanco–Koldobsky–Turnsek theorem says that a linear operator preserving exact Birkhoff–James orthogonality at every point is a scalar multiple of an isometry [2, 3].

## Proof Intuition

For a polygonal unit ball, sufficiently small approximate preservers are already forced by the source's lemmas to respect the finite set of vertex rays. If the polygon has at least six vertices, then it has at least three projective vertex lines. A linear map that sends these finitely many lines to themselves, in some order, is determined up to scalar once the induced line map is fixed; there are only finitely many such possibilities after normalization.

If Property P failed, we could find normalized nonsimilarity preservers with $\varepsilon_k \downarrow 0$. Passing to a subsequence, the finite vertex-ray pattern would stabilize and the operators would converge to an exact orthogonality preserver. The exact Blanco–Koldobsky–Turnsek theorem makes the limit a similarity. But inside a fixed vertex-ray pattern there is only one normalized operator up to sign/scalar, so the whole stabilized sequence was already made of similarities. This contradiction proves Property P.

**Lemma 1** (Finite ray rigidity)**.** *Let $X$ be two-dimensional and let $\mathcal{R}$ be a finite set of at least three one-dimensional subspaces of $X$. Then the set of invertible operators which send every line in $\mathcal{R}$ into a line in $\mathcal{R}$ is a finite union of one-dimensional subspaces of $\mathcal{L}(X)$.*

*Proof.* For each map $\sigma : \mathcal{R} \to \mathcal{R}$, let

$$M_\sigma = \{A \in \mathcal{L}(X) : A(L) \subseteq \sigma(L) \text{ for every } L \in \mathcal{R}\}.$$

There are only finitely many $\sigma$'s, so it is enough to show that whenever $M_\sigma$ contains an invertible operator, $M_\sigma$ is at most one-dimensional.

Choose three distinct lines $L_1, L_2, L_3 \in \mathcal{R}$, with $L_1, L_2$ independent. Since $M_\sigma$ contains an invertible operator, $\sigma(L_1)$ and $\sigma(L_2)$ are independent. Pick nonzero $e_i \in L_i$ and nonzero $u_i \in \sigma(L_i)$ for $i = 1, 2$. After scaling $e_1, e_2$, write $L_3 = \mathbb{R}(e_1 + e_2)$. For $A \in M_\sigma$, there are scalars $a, b$ such that $Ae_1 = au_1$ and $Ae_2 = bu_2$. The extra condition $A(L_3) \subseteq \sigma(L_3)$ says that $au_1 + bu_2$ lies in a fixed line. Because $u_1, u_2$ are independent, this imposes a fixed linear relation between $a$ and $b$. Thus the possible pairs $(a, b)$, and hence the possible operators $A$, form a space of dimension at most one. □

**Lemma 2** (Positive gap for nonsimilar ray maps)**.** *Let $X$ be a two-dimensional polyhedral Banach space whose unit sphere has at least six vertices. Among norm-one operators that map every vertex line of $B_X$ into a vertex line, the nonsimilarities have a positive lower bound on the $\varepsilon$ needed to preserve $\varepsilon$-orthogonality at every point.*

*Proof.* Let $\mathcal{R}$ be the finite set of vertex lines. By the previous lemma, the norm-one ray-preserving operators lie in a finite set, up to sign. Let $\mathcal{F}$ be representatives of those which are not scalar multiples of isometries.

For $A \in \mathcal{F}$, exact Blanco–Koldobsky–Turnsek implies that $A$ does not preserve exact Birkhoff–James orthogonality. Hence there are unit vectors $x, y \in X$ with $x \perp_B y$ but $Ax \not\perp_B Ay$. The Chmielinski–Stypula–Wojcik characterization used in the source gives

$$\min_{f \in J(Ax)} \frac{|f(Ay)|}{\|Ay\|} > 0,$$

because $J(Ax)$ is compact and no member of $J(Ax)$ annihilates $Ay$. Therefore $A$ cannot preserve $\eta_A$-orthogonality for all $\eta_A$ below this positive number. Since $\mathcal{F}$ is finite, the minimum of these positive lower bounds is again positive. □

**Theorem 1** (Classification in polyhedral planes)**.** *Let $X$ be a two-dimensional real polyhedral Banach space. Then $X$ has Property P if and only if the unit sphere of $X$ has at least six extreme points.*

*Proof.* First suppose that $S_X$ has four extreme points. Then $B_X$ is a centrally symmetric quadrilateral, hence $X$ is linearly isometric to $\ell_\infty^2$, equivalently to $\ell_1^2$. The source proves that $\ell_p^n$ fails Property P for $1 \leq p < \infty$; in particular $\ell_1^2$ fails Property P [1, Corollary 2.27]. Thus $X$ fails Property P.

Conversely suppose that $S_X$ has $2m$ extreme points with $m \geq 3$. Let $\mathcal{R}$ be the set of vertex lines. Let $\delta > 0$ be the positive gap from the previous lemma, and choose

$$0 < \varepsilon_0 < \min\{\varepsilon_X, \delta\}.$$

We claim that $\varepsilon_0$ witnesses Property P.

Let $T \in \mathcal{L}(X)$ preserve $\varepsilon_0$-orthogonality at every point. If $T = 0$, then $T$ is a scalar multiple of an isometry. If $T \neq 0$, then the source injectivity lemma makes $T$ invertible. Since $\varepsilon_0 < \varepsilon_X$, the source nonsmooth-point preservation corollary sends each vertex ray into a vertex ray. Thus $\|T\|^{-1}T$ is a norm-one ray-preserving operator. If it were not a scalar multiple of an isometry, the positive gap lemma would force the required preservation constant to be at least $\delta > \varepsilon_0$, contradiction. Hence $\|T\|^{-1}T$, and therefore $T$, is a scalar multiple of an isometry. □

## Verification and Novelty Check

This result is a partial result for the global Property P question, but it is a full classification inside two-dimensional polyhedral spaces. It strictly extends the source's regular polygon theorem: regularity is not used.

Bounded duplicate/literature check performed before promotion:

- local run indexes were searched for arXiv:2501.10719, Property P, approximate preservation of Birkhoff–James orthogonality, and two-dimensional polyhedral variants;
- web search located the source's published Colloquium Mathematicum entry and the later Lebesgue–Bochner continuation arXiv:2601.02786;
- targeted web searches for arbitrary polygon or two-dimensional polyhedral Property P classifications found no exact statement beyond the source's regular polygon theorem.

Human-review recommendation: verify that the source constant $\varepsilon_X$ is strictly positive for every two-dimensional polyhedral space and that [1, Corollary 2.16] is applied exactly as stated. The finite-ray compactness argument is elementary once those source lemmas are accepted.

## References


[1] K. Mandal, J. Manna, K. Paul, and D. Sain, *On approximate preservation of orthogonality and its application to isometries*, arXiv:2501.10719; Colloquium Mathematicum 179 (2025), 189–211.

[2] A. Koldobsky, *Operators preserving orthogonality are isometries*, Proc. Roy. Soc. Edinburgh Sect. A 123 (1993), 835–837.

[3] A. Blanco and A. Turnsek, *On maps that preserve orthogonality in normed spaces*, Proc. Roy. Soc. Edinburgh Sect. A 136 (2006), 709–716.

[4] J. Chmielinski, T. Stypula, and P. Wojcik, *Approximate orthogonality in normed linear spaces and its applications*, Linear Algebra Appl. 531 (2017), 305–317.

[5] Mohit and R. Jain, *Approximate Birkhoff-James orthogonality preserver on Lebesgue-Bochner spaces*, arXiv:2601.02786, 2026.